Chromebook Cart Security System

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Problem Statement

Schools that rely on shared Chromebook carts often struggle with tracking which students have borrowed which devices, leading to lost, stolen, or unaccounted-for Chromebooks. This project addresses that gap by developing an Arduino-based sign-in/sign-out system that allows students to log Chromebook loans using a 12-digit interface and student number, giving administrators a clear, reliable record of device usage and accountability.

Target Audience

This system is designed for high school students and staff in technology-equipped learning environments. Students interact with the system daily to sign out and return Chromebooks, while teachers and administrators benefit from the improved accountability, oversight, and automation it provides. The interface is intentionally simple and familiar, ensuring that users of all technical skill levels can operate it.

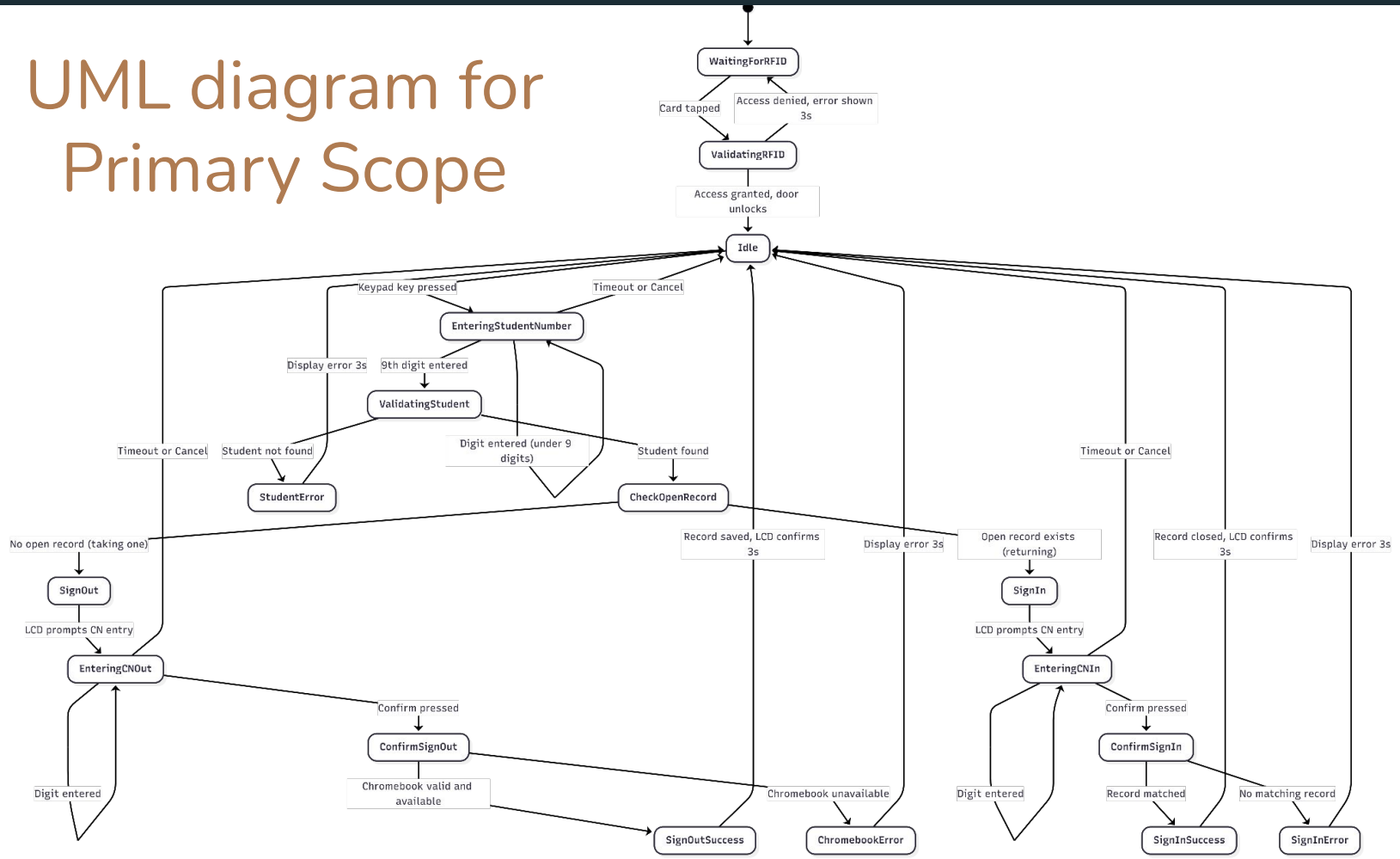
Potential Hurdles

The project presents several technical and logistical challenges, including integrating several hardware components (RFID interface, LCD display, numpad input, various sensors) into a cohesive, reliable system. Ensuring accurate data entry and preventing input errors from students unfamiliar with the interface will also require careful design. Additionally, working within the constraints of the Arduino platform, limited project time, and a three-person team means that scope management will be critical to delivering a functional product.

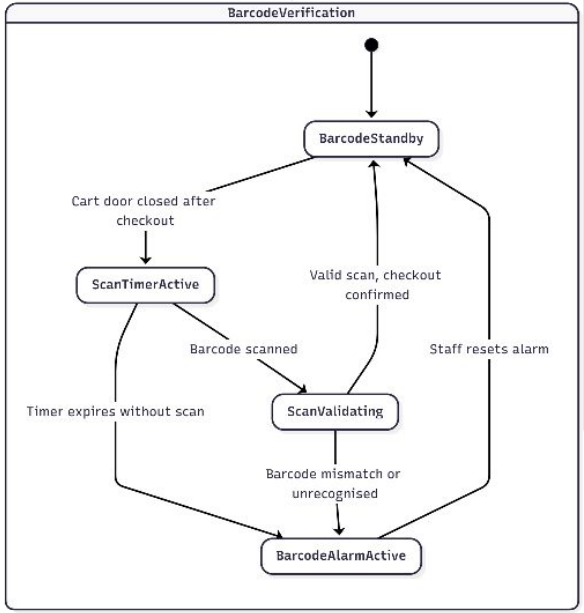
Other ideas

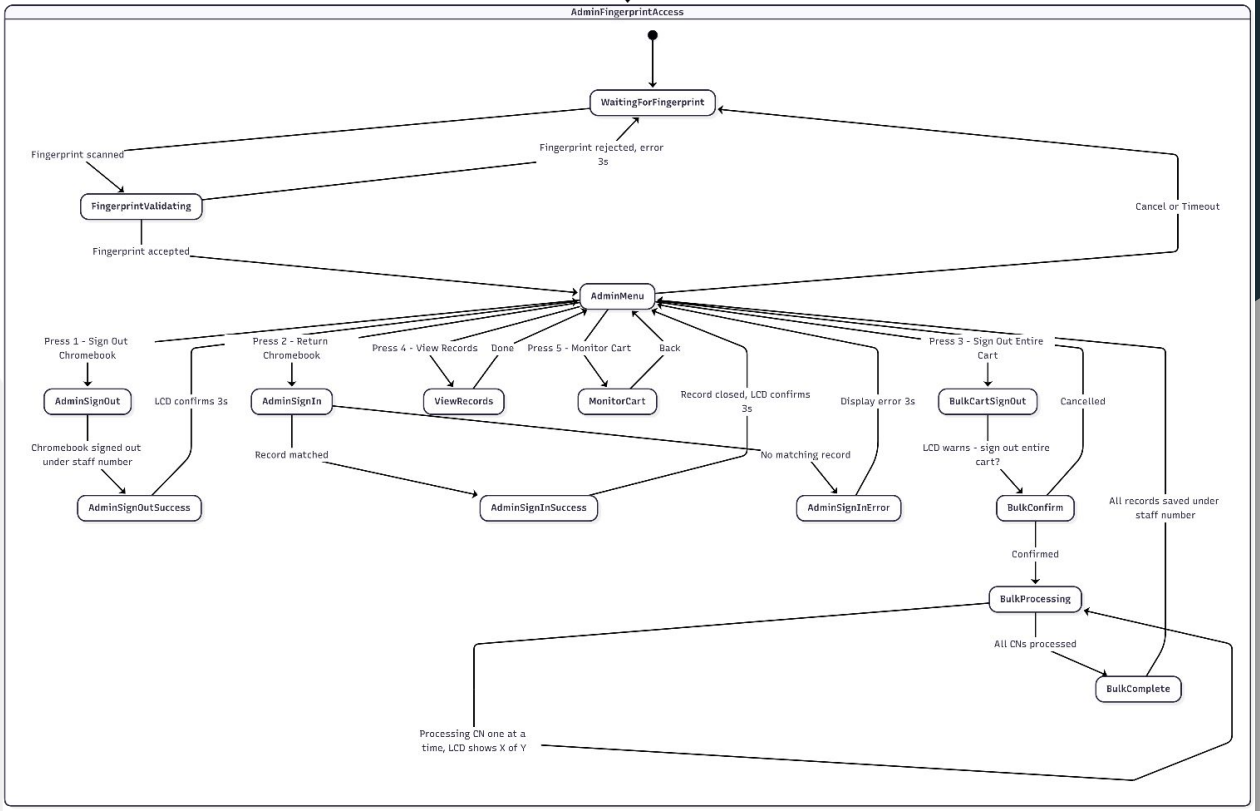
- Library book sign out - Classroom parts list - Digital library sign-in sheets - Room focus meter based on noise input (For working periods, library) - Shop dust & air quality sensor	We wanted to make use of more interesting components such as a fingerprint sensor or RFID scanner. However we also wanted to mainly use materials that we already had to avoid purchasing too much. Additionally, we wanted our idea to be unique with easy ways to expand upon the base idea.

UML diagram for Primary Scope

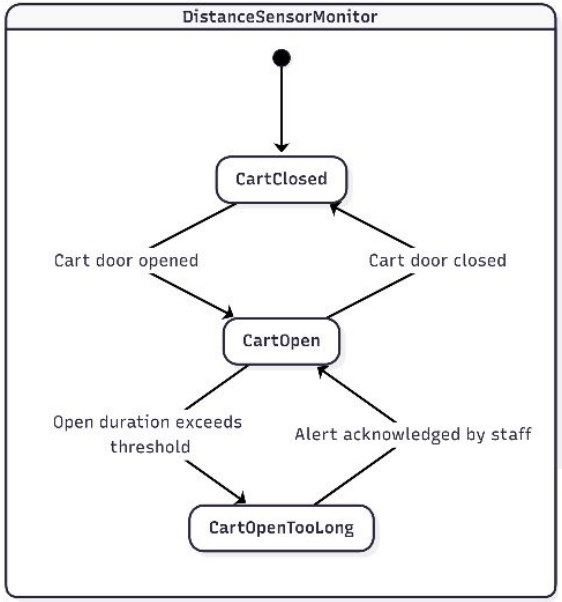


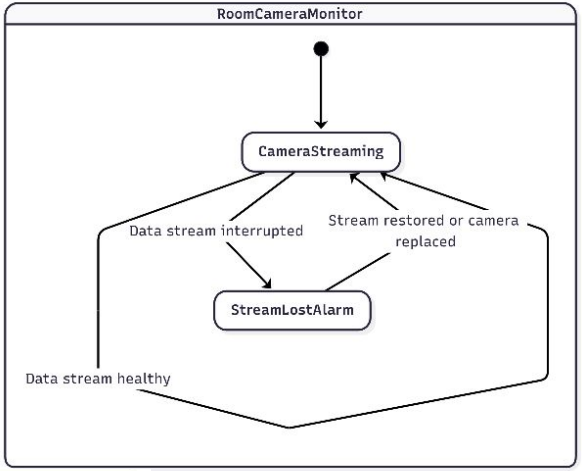
UML diagram for Secondary Scope

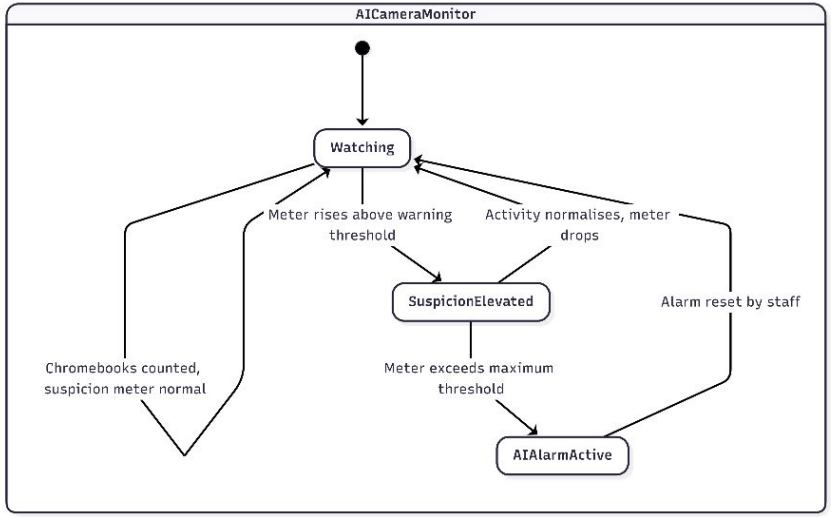




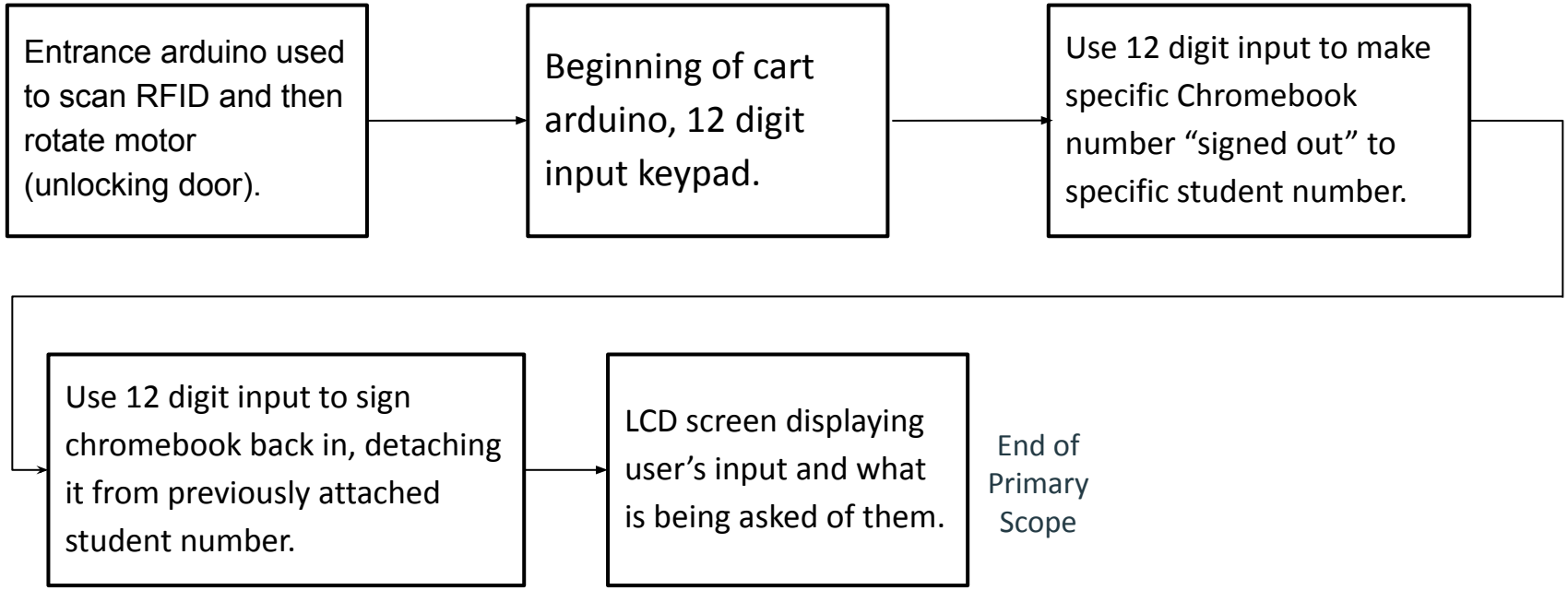
UML diagram for Tertiary Scope







Roadmap (Primary Scope)



Roadmap (Secondary & Tertiary)

Barcode reader to scan Chromebook, trigger a buzzer if it takes too long to receive input.

Enter Chromebook cart with a fingerprint sensor for admin access. Admin access gives access to signing out an entire cart.

End of Secondary Scope

DIstance sensor to check if the cart was opened without authorization.

A camera inside the Chromebook cart that uses AI to count the number of Chromebooks being taken out. A “suspicious” meter will be used. If the meter is past a certain threshold, an alarm will sound.

A camera in the room, simply for security purposes, to review footage. (Note: ceiling is high, camera is hard to dismantle). If the camera stops sending data, an alarm sounds.

End of Tertiary Scope

Feedback

Othman Boudjelthia	Due to student numbers being easily accessible, some sort of authentication system should be involved to confirm students.
Marcus Leanage	What protection does your system have from students taking more than one chromebook while signing them out?
Summer Morrison	A camera should be used to monitor the room and account for any unforeseen concerns.
Ahmed Hussein	What protection does your system have from students disconnecting the power to disable the security measures?
John Wang	Your current system will be too difficult to integrate into existing chromebook carts.

Takeaway

The biggest takeaway from the feedback is the fact that while these ideas do solve certain issues related to chromebook security, the system itself has its on flaws and gaps within its safeguard systems. Issues such as people not hearing the alarm or students taking out more than one chromebook are issues that currently have a small scale solution but would be difficult and tedious to administer on a large scale. The amount of added security required to stop issues like that would be very difficult to implement into the current chromebook carts and would probably require a full overhaul of them.

Work Cited

Claude.ai prompt used:
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Write a problem statement for my project:

Arduino 3-person project for TEJ4M. Primary Scope: Enter the Chromebook cart room with an RFID 9-digit interface on Chromebook carts: Enter student number and Chromebook number Use the 9-digit interface to sign out of the Chromebook with SN and CN Use a 9-digit interface to sign in to Chromebook as well Each Chromebook cart contains an LCD with instructions (above the 9-digit interface) Secondary Scope: Barcode scanner tells which computer to take based on the given barcode Enter Chromebook cart with a fingerprint sensor for admin access Tertiary Scope: Blacklist individuals (based on student number) who took too long to return the Chromebook A distance sensor inside the Chromebook cart that monitors if the cart is open.

Write 1 short and concise small paragraph
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UML DIAGRAMS ORGANIZED BY SCOPE (with assistance from mermaid.ai):
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